SVKM's Narsee Monjee Institute of Management Studies
Mukesh Patel School of Technology Management & Engineering

Program: B Tech / MBA Tech (All Programs except for AI, AIML, AIDS, Civil and Mechanical)	**Semester:** II
Course/Module: Quantum Physics	**Module Code:**

Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks- 100 in Question Paper)**
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50

Course Objectives:

The objective of this course is to teach the limitations of classical physics in explaining several experimental facts leading to the development of quantum physics. Also, this course will expose the students to different applications of quantum mechanics, such as quantum tunneling and quantum computation.

Course Outcomes:

After completion of the course, students would be able to:

1. Understand the need for Quantum mechanics (K2)
2. Explain the basic laws related to quantum mechanics and apply them to solve related problems. (K3)
3. Apply the concept of quantum mechanics to real-life instrumentation and applications. (K3)
4. Understand the requirement of quantum computation and its various aspects. (K2)
5. Gauge the theoretical principles through laboratory experimentation, enhancing problem-solving abilities and technical proficiency in modern physics applications. (K3)

Detailed Syllabus:

Unit	Description	Duration
1.	**Limitations of classical physics and introduction to old quantum theory:** Introduction to Quantum Physics, Black body radiation, Explanation of it using the photon concept, Photoelectric effect, Compton effect, de Broglie hypothesis, Experiments demonstrating wave properties of electron: Electron interference (double slit experiment), Electron Diffraction (Davison - Germer experiment), Uncertainty Principle, Verification of matter waves.	**6**
2.	**Schrodinger Equation and its applications**: Basic postulates of quantum mechanics, concept of wave function, Born's interpretation of the wave function, Superposition principle of eigenstates. Concept of collapse of wave function. Time-dependent and time-independent Schrodinger Equation.	**6**
	Concept of the free particle, particle in an infinite and finite potential well, box problem. Bound vs. unbound states.	**2**

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3.	**Concept of Quantum Tunnelling and its applications**: Concept of Quantum Tunnelling. Reflection and Transmission co-efficients. Few realistic examples of tunnelling, e.g., alpha decay, Probe microscopes (Scanning Tunnelling microscope).	**5**
	Simple Harmonic Oscillator, explanation in 1D (no detailed derivation). Hydrogen atom.	**3**
4.	**Introduction to Quantum Computing**: Limitations of classical computation, Difference between classical and quantum computer, Qubits, Bloch Sphere, Single Qubit Operations.	**5**
	Quantum Gates, Bell States, Quantum Entanglement.	**3**
	Total	**30**

Text Books

1. A. Beiser, S. Mahajan and S. Choudhury, "Concept of Modern Physics", Tata McGraw Hill, 7th Edition (SIE) 2015.
2. Arthur Beiser, Perspectives of Modern Physics, McGraw Hill, 1969.
3. Bernhardt, Chris, Quantum computing for everyone, MIT Press, 2020.

Reference Books

1. Eisberg and Resnik, "Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles" Wiley, 2nd edition 2006.
2. R. A. Serwey, C. J. Moses, C. A. Moyer, "Modern Physics", Thomson, 3rd edition 2005.
3. David J. Griffiths, "Introduction to Quantum Mechanics", Pearson, 2nd Edition 2015.
4. Frederick Reif, "Fundamentals of Statistical and Thermal Physics", Waveland press 2009.
5. McMahon, David, Quantum computing explained, John Wiley & Sons, 2007.

Laboratory Work

Minimum 8 to 10 practical exercises based on the syllabus.

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